



OFFICIAL

Arterial Line Management Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All services	Medical and Nursing

2. Purpose

An arterial cannula may be inserted via the radial, brachial, femoral or dorsalis pedis arteries for the purpose of continuous monitoring of systemic arterial pressure and provide ready access for sampling of arterial blood.

Monitoring Indications for use may include

- Haemodynamic instability
- To guide titration of vasoactive medications (e.g. Noradrenaline)
- Major surgical intervention
- Respiratory failure

3. Procedure

3.1 Arterial Line Insertion

3.1.1 Key points prior to commencing any procedure

- Explain the procedure/s to the patient, family and/or carer and gain appropriate consent.
- Consideration should be given to vascular integrity prior to arterial cannula insertion due to a potential compromise to distal circulation.

3.1.2 Equipment Required

- Equipment must be appropriate for the age and/or size of the patient.
- Equipment must be serviced and calibrated in accordance with manufacturer's recommendations to ensure reliability and accuracy.

3.1.3 Patient Preparation

- Explain the procedure to patient/relatives.
- Verbal consent should be obtained by the Medical Officer (MO) performing the procedure.
- The Radial and Femoral arteries are typically cannulated for arterial monitoring, however the Brachial and Dorsalis Pedis can also be utilised.
- The preferred site for arterial monitoring is the radial artery.

Before referencing this procedure document, please ensure that you have the latest version from the [Policy Library](#) information hub page.

- The MO performing the procedure should do an Allen's Test to ensure adequate distal blood flow (see Figure 1).

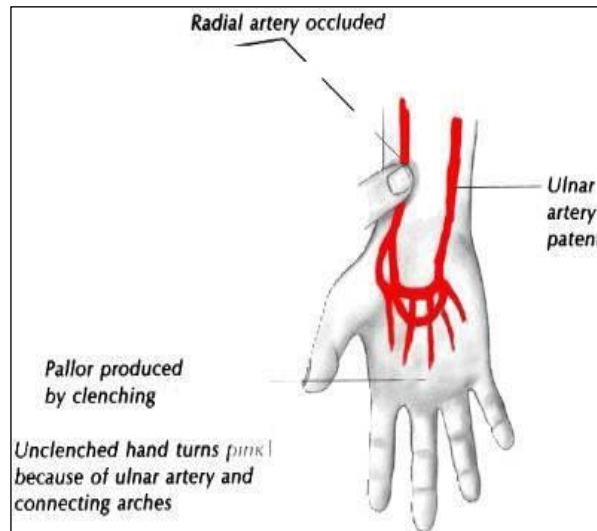


Figure 1.

- Position the patient in bed
- If a radial artery is to be used, a rolled towel may be used to hyperextend the wrist to allow easier visualisation of landmarks. (see Figure 2).
- Brachial insertion - elbow extension.
- Femoral insertion - lay patient supine if possible.

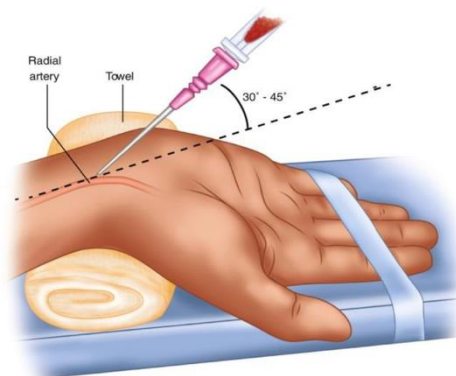


Figure 2.

3.1.4 Transducer Setup

- Assemble the pressure monitoring set.
- Open the packet **and tighten all connections first**.
- Attach transducer set to a 500mls bag of Sodium Chloride 0.9%.
- Insert Sodium Chloride 0.9% bag into a pressure infuser bag.
- Pressurise the infuser bag to 300mmHg as indicated on pressure bag.
- Ensure that **all air bubbles are removed from the transducer set / flush system** and that all tubing has been primed with fluid.

- Ensure all assembled equipment is ready for access and use prior to line insertion
- Sodium Chloride 0.9% 500mls is used for line patency and flushing.
- Ensure all priming caps are replaced with those provided by manufacturer and connections are secured.

3.1.5 Cannula Insertion

- Explain procedure to the patient/parent; obtain patient's consent.
- Wipe the dressing trolley with alcohol wipes.
- Prepare dressing trolley with the required equipment.
- Check patient medical chart/record for allergies, physician orders, site restriction and potential bleeding conditions.
- Position the patient so that the hand is dorsiflexed at the wrist approximately 60 degrees and palpate the artery.
- Wash hands.
- Administer local anaesthesia.
- Open arterial cannula insertion kit and then don sterile gloves.
- Cleanse the skin with chlorhexidine.
- Create a sterile field around insertion site utilising appropriate drapes.
- Insert the cannula at a 30-45-degree angle.
- Plug the primed transducer into the monitor and attach the arterial cannula to the transducer line. Ensure that an identifiable arterial trace and blood pressure reading is present prior to proceeding.
- Suture to keep the cannula secure and apply a transparent/ CHG dressing -apply dressing after attaching transducer cable & ensuring arterial trace. 3M Cavilon, no sting barrier film to protect the skin and also to get a good seal with dressing.
- The arterial line site must always be visible to detect any bleeding. Do not obscure with pillows, sheets, blankets etc.

3.1.6 Zeroing of the Arterial Line

This should take place:

- Following insertion of a new Cannula with initial transducer connection.
- Following line changes.
- Following any transducer disconnection.
- Upon suspected erroneous readings on the monitor.
- At the start of each shift/once every shift.

How to zero the Arterial Line:

- Turn the three-way tap most distal to the insertion site or near the transducer 'Off' toward the patient (See Figure 3 below)

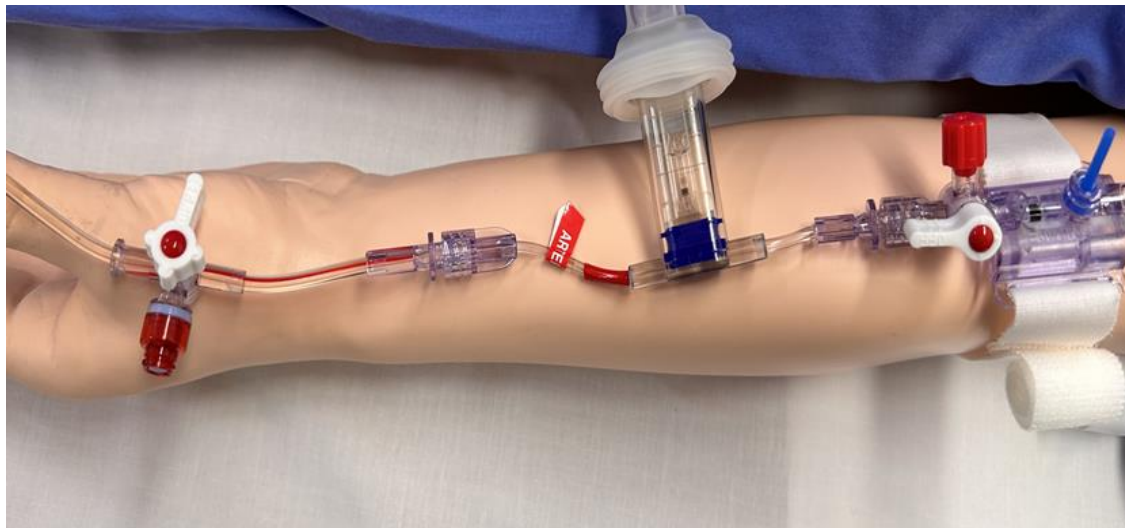


Figure 3.

- Remove the cap to open the transducer line to air. (See Figure 4 Below)

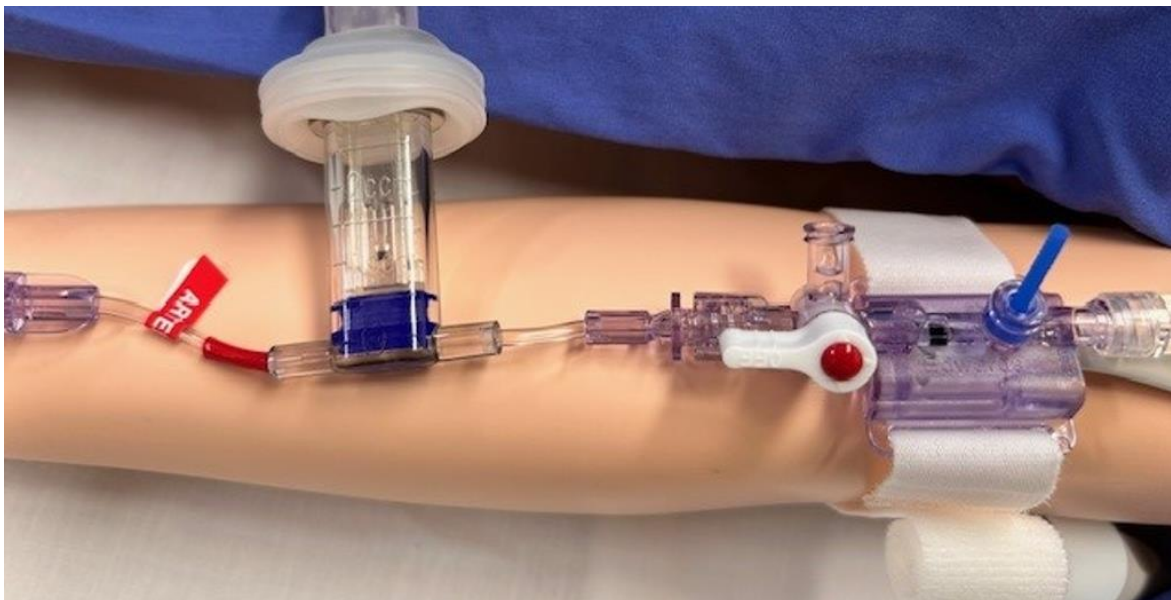


Figure 4.

- Press the Zero ART button on the monitor.
- When the monitor reads zero and shows a flat trace line, close the three-way tap and replace the bung.
- When the transducer is secured to a transducer mount at the end of the patients' bed, the transducer must be zeroed at the level of the phlebostatic axis. This site is at the intersection of the fourth intercostal space and mid axillary line (See Figure 5.).
- Transducer mount is secured over patient's arm

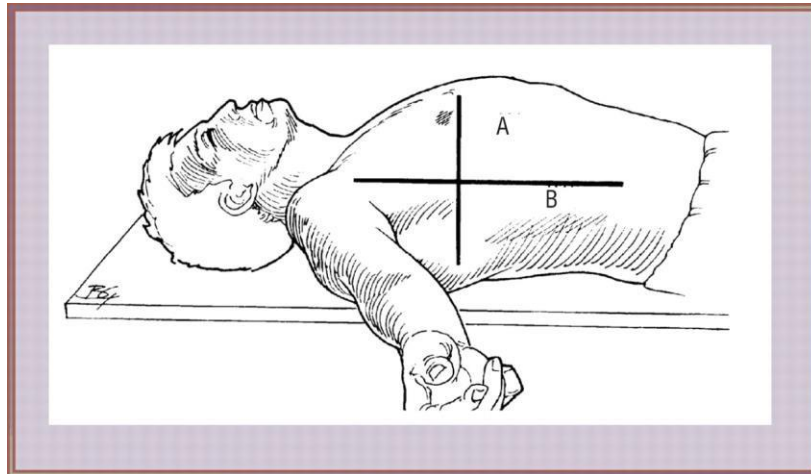


Figure 5

3.1.7 Patient Monitoring/Management

- Once the arterial Cannula has been secured with a suture, apply the transducer dressing to the cannula over the insertion site.
- All lines connected to the transducer and related to the arterial line should be labelled red as per [National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines \(safetyandquality.gov.au\)](http://www.safetyandquality.gov.au).
- Ensure that the flush line is secured to patient with velcro strap from transducer kit.
- Document in patients notes and on the Nursing care plan the insertion date, site and type of dressing.

Note:

The insertion site and limb distal to the insertion site should always remain in sight, and checked at regular intervals for bleeding, warmth, sensation, colour and pulses if applicable
Arterial Pressure Alarms MUST NEVER BE TURNED OFF.

Clinical observations should be in line with prescribed parameters defined for each patient and at a minimum should include:

- Recording of transduced Blood pressure hourly.
- Regular neurovascular assessment of cannulated limb. Every 4 hourly or as directed by MO if there are issues.
- 4 hourly Temperature recording of the limb, where the A line is situ.
- Observe the limb, distal to the insertion site for warmth, sensation, colour, and pulses if applicable. Document in the progress notes.
- The setting of clinically appropriate alarms.
- Maintain flush bag pressure 300 mmHg.
- Arterial wrist splint can be applied if necessary to maintain positioning.

If there are doubts about the accuracy of the arterial line, perform and record a non-invasive blood pressure. Escalate any concerns to the medical team.

3.2 Transducer Line Changes

Performed at 96hrs or if faulty.

3.2.1 Equipment

- Gloves [Nonsterile].
- Safety glasses.
- Infusion pressure bag.
- Sodium Chloride 0.9% 500mls.
- Arterial transducer set.
- Dressing pack.
- Antiseptic solution (e.g. Chlorhexidine 2%/Ethanol 70%).
- Clear dressing (e.g. Tegaderm CHG or Opsite™ 3000).

3.2.2 Procedure

This is a two-person procedure if the arterial cannula does not have a flow-switch.

- Don safety equipment and silence monitor alarms. Explain the procedure to the patient as necessary.
- Spike the 500mL bag of sodium chloride 0.9% with the pressure transducer line, the line should be fully primed and then inflate the pressure bag to 300 mmHg.
- With assistance, remove dressing from the arterial line site; observe for excessive bleeding, inflammation or infection.
- Clean the area with antiseptic solution and allow to dry.
- (If flow switch is not present) Ask a colleague to occlude the artery proximal to the insertion site. Note: If occlusion is effective, the waveform will be flattened. Occluding the artery is not required if the arterial cannula has a flow-switch that can be pushed into the "OFF" position to prevent retrograde blood flow.
- Unscrew the flush line and remove. Replace immediately with clean line.
- Ensure a colleague supports the new line and Cannula to avoid dislodgement.
- Ensure transducer is attached to pressure cable and monitor, and that there is an identifiable arterial trace.
- Apply the clear dressing to Cannula over the insertion site.
- Ensure flush line is secure to patient with Velcro strap from transducer kit.
- Document in the patient notes, and on the Nursing Care Plan, the line change and dressing type. Dressings should be changed every seven days, or sooner if visibly soiled.
- Observe the limb, distal to the insertion site for warmth, sensation, colour, and pulses if applicable.
- The transducer line is to be changed every 96 hours/PRN. This is to include all lines from the cannula to the continuous flush device.¹

3.3 Blood Sample Collection

Note: the use of closed-circuit transducer sets for the collection of blood is recommended as best practice; however, it is not used in all areas within AKG.

3.3.1 Equipment

- Gloves.
- safety glasses/goggles.
- 5mL syringe.
- Required blood sample tubes and arterial blood gas (ABG) syringe if needed.

- sterile gauze swabs.
- alcohol swabs.
- Arterial line vacutainer adaptor/blood transfer device.

3.3.2 Procedure

1. Don safety equipment, silence monitor alarms and explain procedure to patient as necessary. Turn the three-way tap most distal to the insertion site toward the patient (See Figure 6 below.)



Figure 6.

2. Firmly squeeze the flexures and slowly (over 3-5 seconds), pull up the reservoir plunger until the reservoir has reached its **3ml** volume capacity (see Figure 7 below.). Make sure there is no saline present in the blood sample.

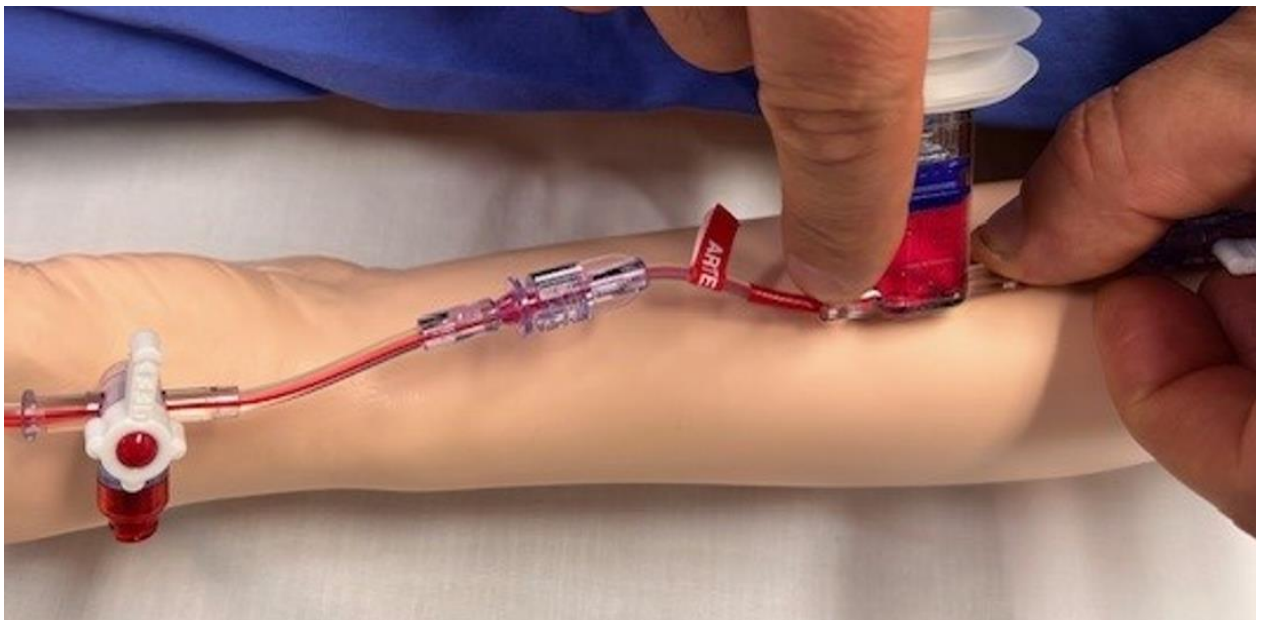


Figure 7.

3. Turn the three-way tap closest to the 'insertion site' toward the transducer. This 'locks' the blood mixed with saline between the sample port and the transducer. Swab the sample port. Push the syringe or vacutainer adaptor into the sample port. (see Figure 8 below)

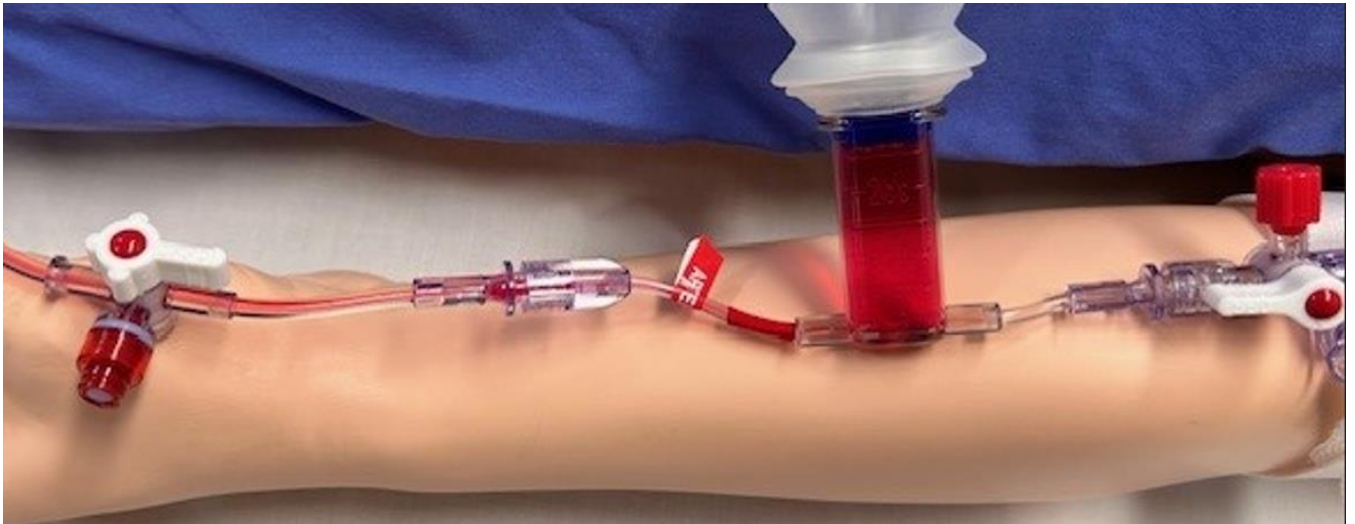


Figure 8.

4. Firmly insert the blood sample tube/s to the arterial line three-way port (Note: If blue (citrate tube) is required collect this first. **Collect the ABG last.** (See Figure 9 below.). You may need to manipulate the position of the patient's wrist to draw blood if the arterial line is positional.

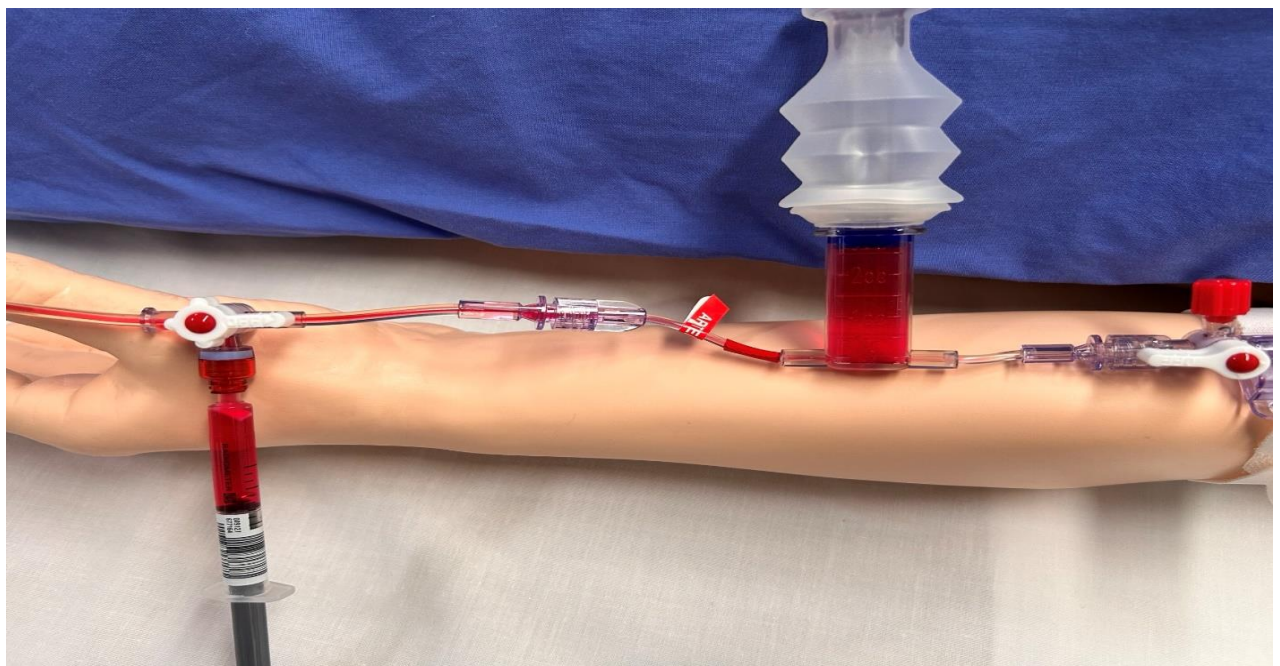


Figure 9.

5. After the last sample has been drawn, remove the syringe or vacutainer adaptor.
6. Now the blood can be returned from the plunger to the patient. To do this, turn the three-way tap **closest to the insertion site 'OFF' to the sampling port to prevent blood squirting out** from sample port when returning blood with pressure. (See Figure 10 below)

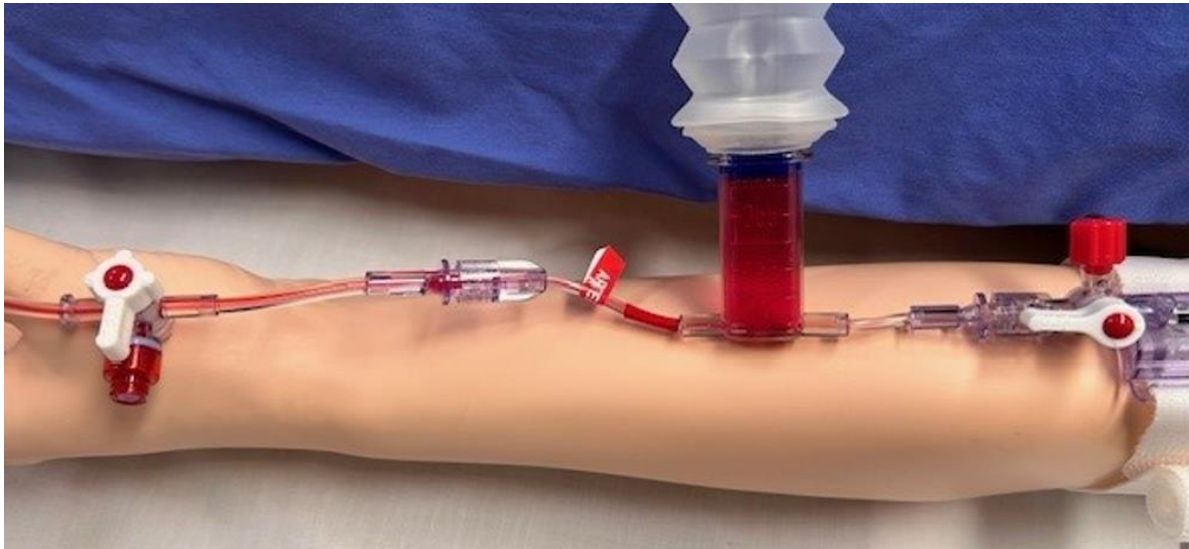


Figure 10.

7. Now you can push the blood slowly from the plunger and return to the patient. Smoothly and evenly over 3-5 seconds, push down on the plunger until the flexures lock in place in the fully closed position and all the fluid has been reinfused into the line. (See Figure 11 below)

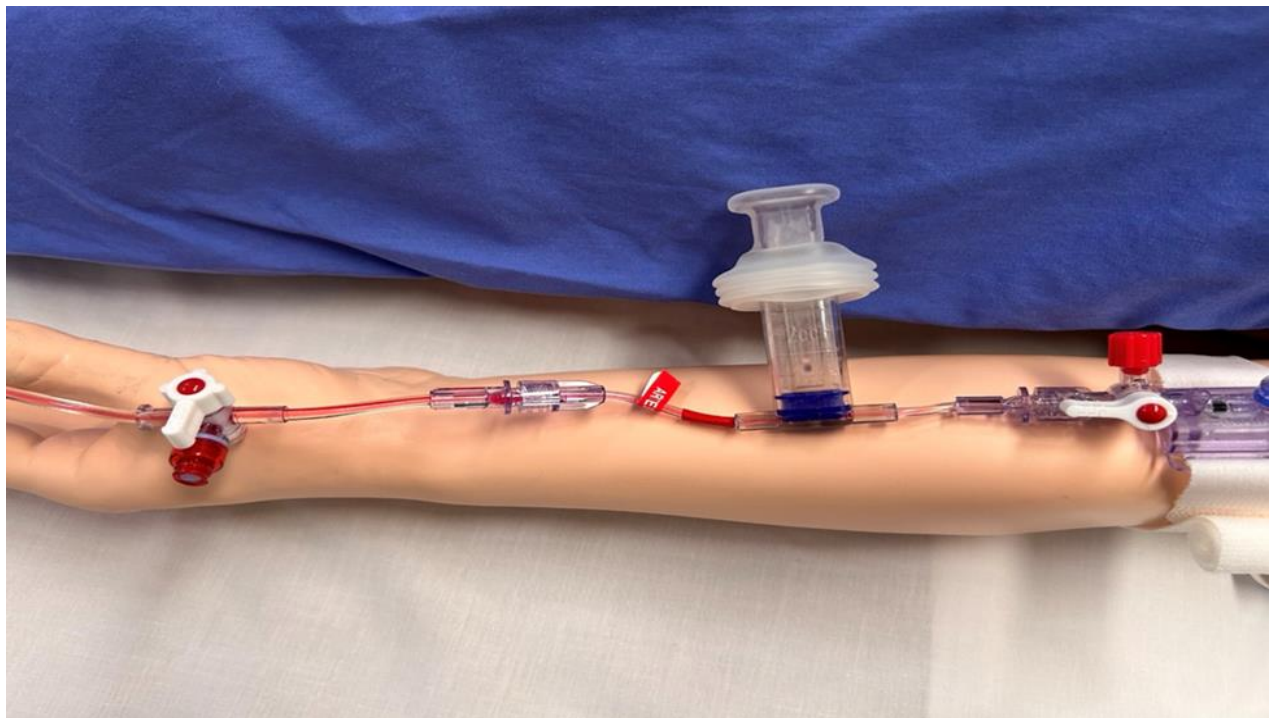


Figure 11.

8. After returning the blood, keep the three-way tap in the opposite direction to the sample port. Turn until it is in the six o'clock position. Then only the three-way sampling port will get flushed by saline and clear any blood from the sampling port. (See Figure 12 below).

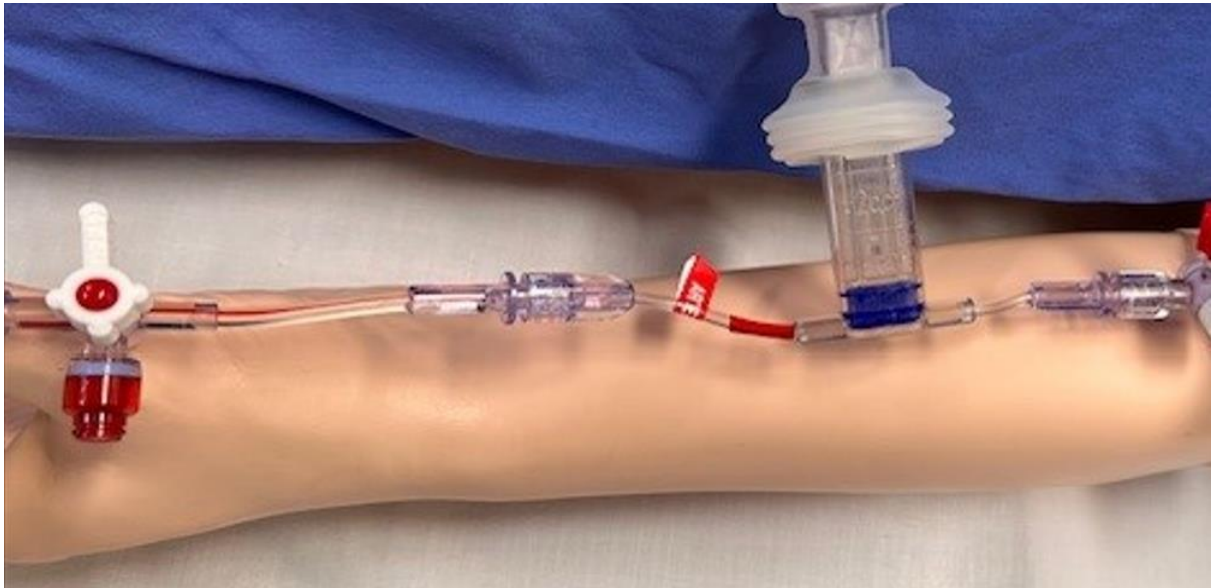


Figure 12.

9. Open both the three-way taps proximal & distal to the patient to flush. Flush the VAMP system clear by pulling the 'snap-tab' on the TruWave transducer (See Figure 13 below).

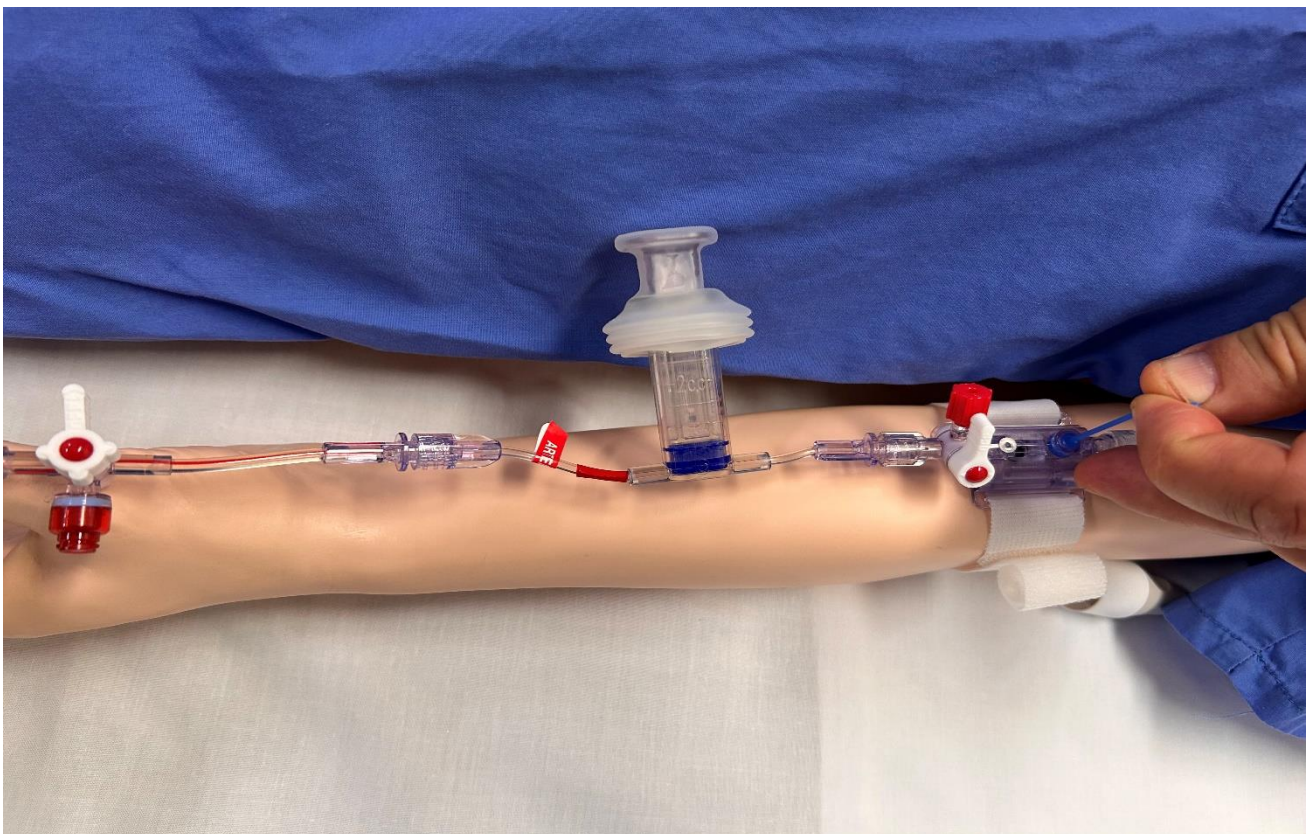


Figure 13.

10. Next is the clearing of the A line from any blood or clots. For this procedure, first close the **three-way tap to the insertion site**. This will prevent blood coming back to the line. Attach a 10ml syringe to the sample port to aspirate/ flush any residual blood in the line. (See figure 14 below).

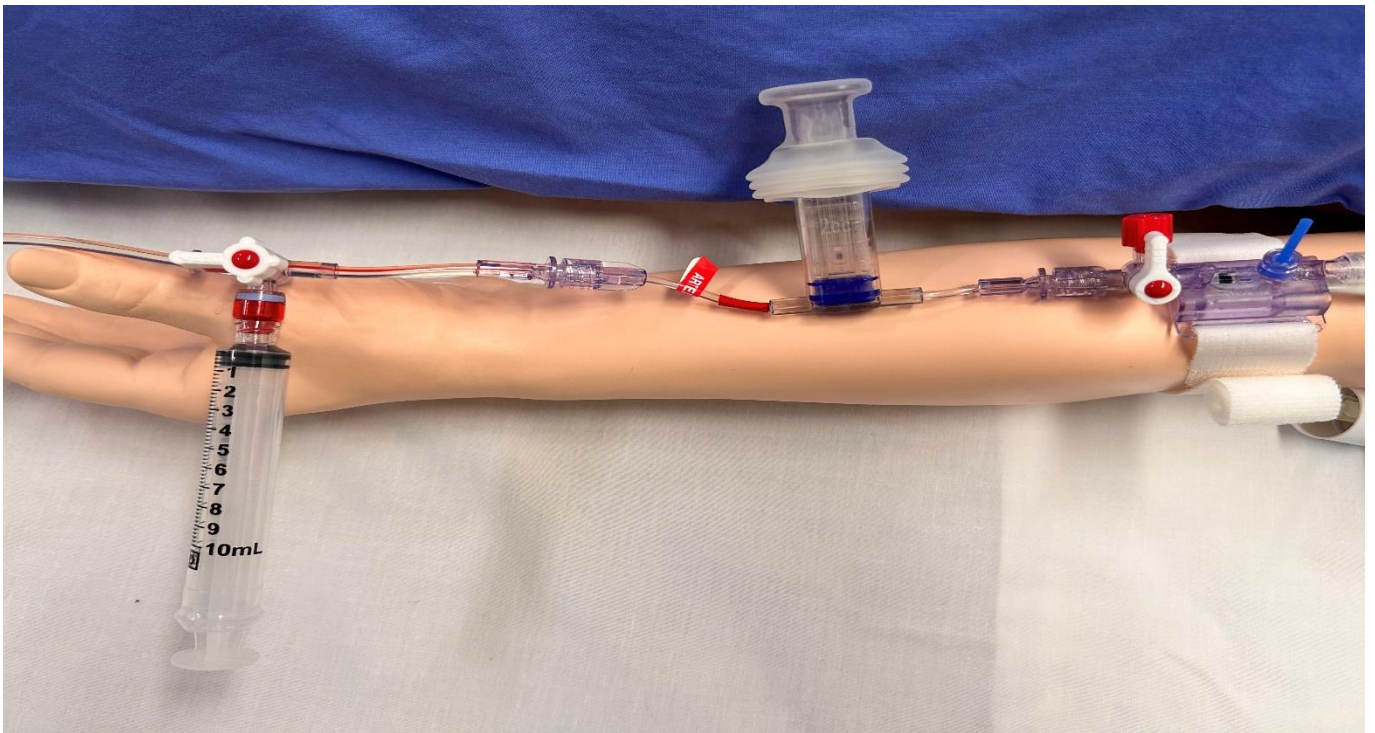


Figure 14.

11. For the flushing process, pull the 'snap-tab' on the transducer.

- Collect saline in the reservoir by pulling the 'snap-tab' and pulling the reservoir plunger at the same time. (See Figure 15 below).

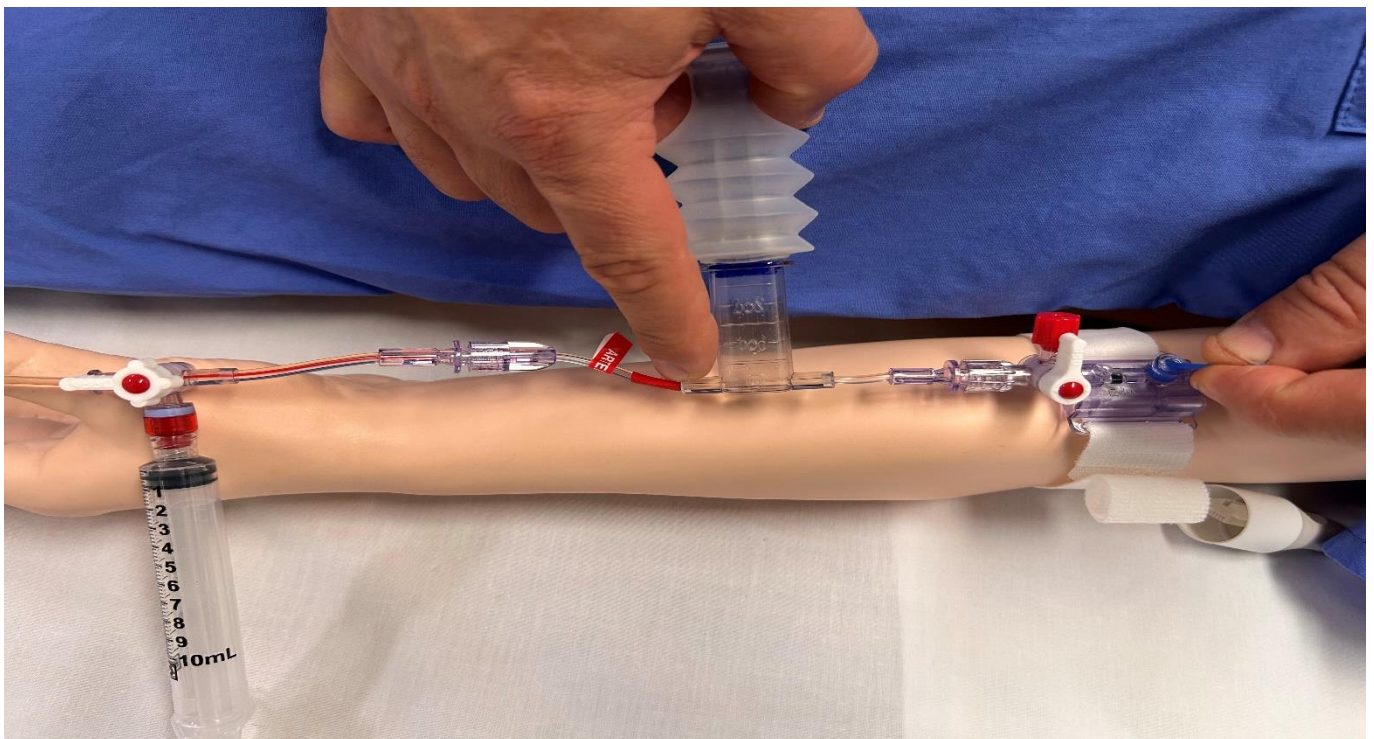


Figure 15.

- Close the three-way tap **near the transducer off towards the patient.**
- Then **use the plunger to flush the saline slowly using 'push- pause' technique to the attached 10ml syringe.** This process makes sure any leftover blood from the three-way port is cleared. (See figure 16 below).

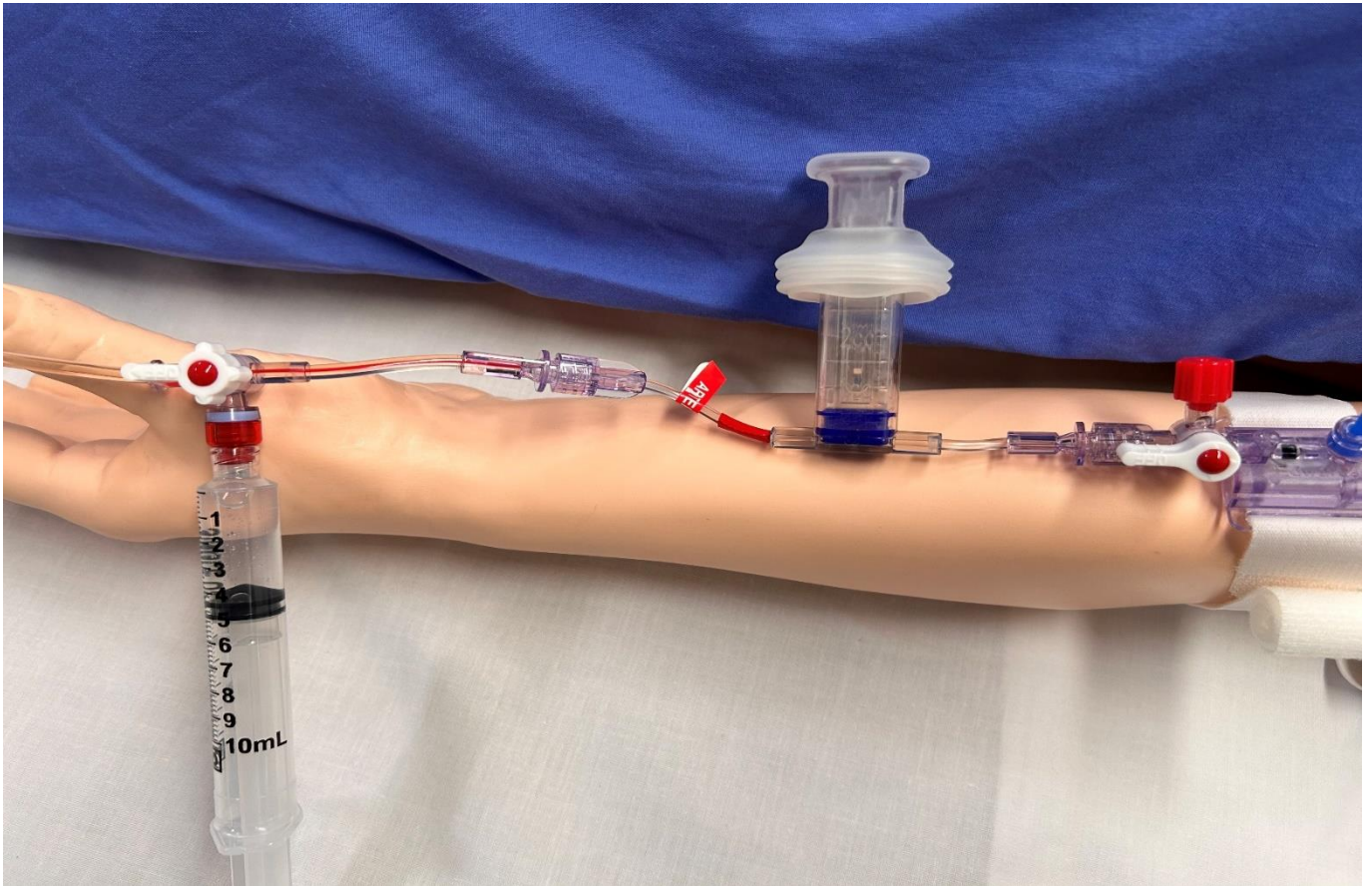


Figure 16.

12. Remove the 10 ml syringe and open all three-way taps to allow for arterial blood pressure monitoring again. Final position of three ways shown in figure 17 below.

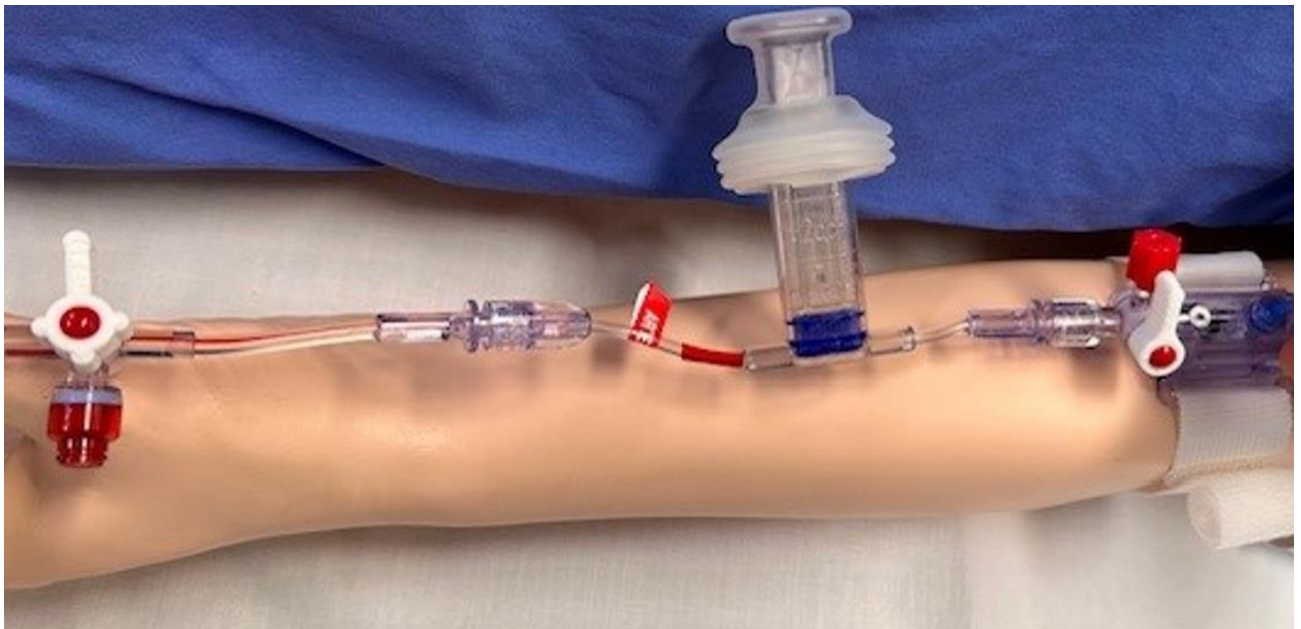


Figure17.

3.4 Removal of Arterial Line

It is not recommended that arterial Cannulas are routinely replaced, however when they are no longer required, they should be promptly removed.

3.4.1 Equipment

- Gloves Safety glasses/goggles.
- Dressing pack.
- Sterile gauze swabs.
- Stitch cutter.
- Antiseptic solution.
- Adhesive tape. Pressure dressing
- Specimen container *
- Sterile scissors *

3.4.2 Procedure

- Check the patient's recent coagulation results if clinically indicated.
- Ensure the patient is comfortable and the arterial line is easily accessible. Explain the procedure as necessary.
- Don safety equipment and silence the alarms.
- Turn three-way tap distal to the insertion site off to the patient, deflate the pressure bag.
- Remove the dressing from arterial cannula site.
- Clean site with antiseptic solution.
- Cut suture if present.
- Whilst applying firm pressure to the entry site with sterile gauze swab, remove arterial cannula and place on second sterile gauze swab.
- Maintain firm pressure **for five minutes, or until bleeding stops.**
- Apply a third sterile gauze swab as a dressing and firmly cover with adhesive tape/fixomul. Try to use a pressure dressing with fixomul in the beginning and revisit the site after 15 to 30 minutes. If there is no bleeding remove pressure and change to a normal dressing.
- Ensure the insertion site always remains visible and monitor for haematoma or signs of bleeding.
- Document removal on Nursing Care Plan and patient notes.

***If required**, aseptically preserve the Cannula tip by cutting with sterile scissors and placing in specimen container and send to microbiology

4. Legislative context

The following documents are required to give effect to this procedure:

- HDWA
 - [Consent to Treatment Policy](#)
- EMHS
 - [Infection Prevention and Management Policy EMHS: 34](#)
 - [Consent Policy EMHS: 5](#)
- AKG
 - [Hand Hygiene Policy](#)

5. Supporting information

The following documents support the implementation of this procedure:

Arterial Line Management Procedure

Version: 4 | Last Reviewed: December 2025 | Next Review Date: December 2028

Page 13 of 18

- AKG
 - [Aseptic Technique Procedure](#)
 - [Specimen Collection and Pathology Results Procedure](#)
- Other
 - [National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines \(safetyandquality.gov.au\).](#)

6. Compliance, monitoring and evaluation

Compliance against this guideline will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the department meeting for review and identification of any opportunities for improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3: The health service organisation uses a charter of rights that is:
 - a. Consistent with the Australian Charter of Healthcare Rights
 - b. Easily accessible for patients, carers, families and consumers
- Standard 4 – Action 4.3: Clinicians use organisational processes from the Partnering with Consumers Standard in medication management to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 5 – Action 6.5: The health service organisation:
 - a. Defines approved identifiers for patients according to best-practice guidelines
 - b. Requires at least three approved identifiers on registration and admission; when care, medication, therapy and other services are provided; and when clinical handover, transfer or discharge documentation is generated

8. Policy owner (relating to these procedures)

Head of Department Intensive Care Unit

Enquiries relating to this procedure may be directed to:

Title: Head of Department
Department: Intensive Care Unit
Email: Reyas.aboobackerKaniyamparambil@health.wa.gov.au

9. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	04 12 2025	04 12 2025	Full review; Significant changes throughout document

10. Authorisation

This procedure has been authorised and issued by:

Authorised by	Dr Tania Strickland - Director Clinical Services
Authorisation date	04 12 2025
Published date	04 12 2025

11. Appendices

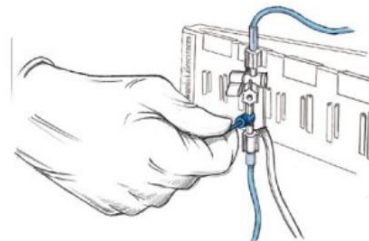
11.1 Appendix 1 – Priming Closed-circuit transducer line – Used in ICU



VAMP Closed Blood Sampling System Venous Arterial blood Management Protection system Quick Reference Guide

Priming

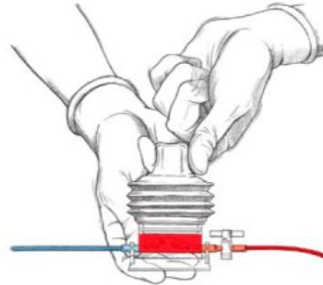
1. Squeeze and hold flexures to slightly raise plunger to facilitate flow of priming solution.
2. With the shut-off valve in the open position (parallel to the tubing), hold sample site above the VAMP reservoir at a 45° angle.
3. Provide flow by pulling the Snap-Tab of the TruWave disposable pressure transducer.
4. Slowly deliver priming solution to remove air. Close plunger and connect to patient's catheter.



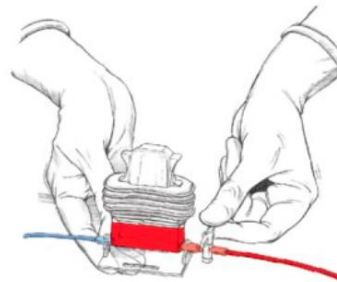
11.2 Appendix 2 – Sampling from Cardiac output monitoring line– in ICU

Drawing the clearing volume

5. Firmly squeeze the flexures and slowly (over 3-5 seconds) pull up the reservoir plunger until the reservoir has reached its 5 mL volume capacity.

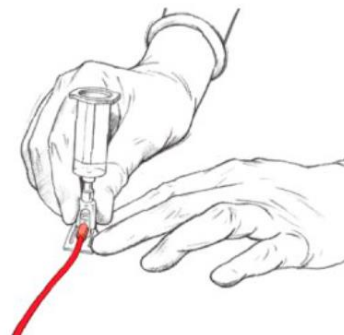


6. Close shut-off valve by turning the handle perpendicular to tubing.

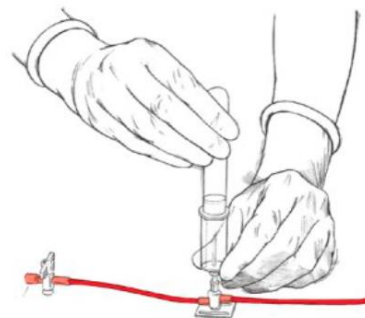


Drawing blood samples from Z-site needleless sample site

7. Swab the sample site. Push cannula with syringe or direct-draw unit into sample site.

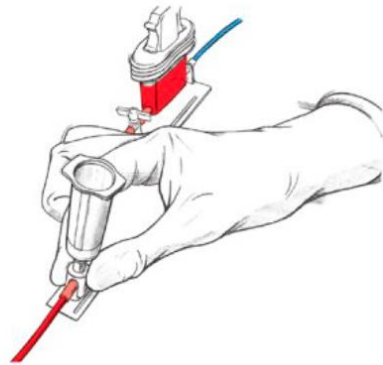


8. If using direct-draw unit, put vacuum tube into the unit. Draw the required volume of blood for a sample. Repeat to complete blood study requirements.

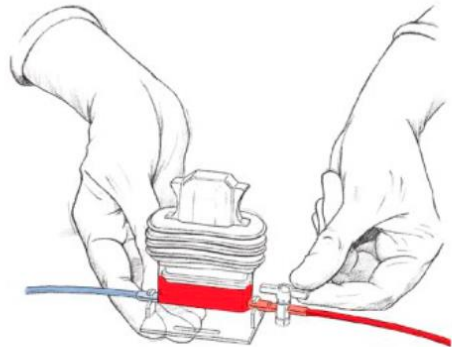




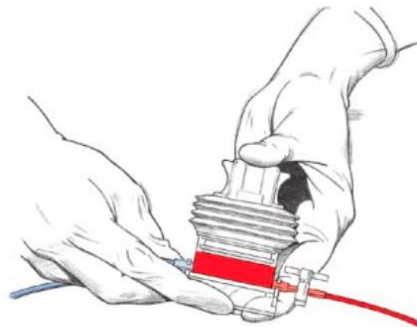
9. After the last sample has been drawn, remove the vacuum tube first, then grasp the VAMP direct-draw unit by the cannula and pull straight out.



10. Open shut-off valve by turning the handle parallel to tubing.



11. Smoothly and evenly over 3-5 seconds, push down on the plunger until the flexures lock in place in the fully closed position and all fluid has been reinfused into the line.



12. Flush the VAMP system clear by pulling the Snap-Tab on the TruWave transducer and swab the sample site ensuring removal of any excess blood.

